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OPTICAL IMAGING SYSTEM

Abstract

An optical imaging system includes: a first lens group including a plurality of lenses; a second lens group including a plurality of lenses; a first reflecting member disposed on an object side of the first lens group; and a plurality of second reflecting members arranged between the second lens group and an image sensor. One or both of the first lens group and the second lens group is movable along an optical axis, an interval along the optical axis between the first lens group and the second lens group is varied between a first interval and a second interval smaller than the first interval, and as at least one of the first lens group and the second lens group is moved, one or more of the second reflecting members are moved.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION(S) [0001] This application is a continuation of U.S. application Ser. No. 18/428,799 filed on Jan. 31, 2024, which is a continuation of U.S. application Ser. No. 17/215,699 filed on Mar. 29, 2021, now U.S. Pat. No. 11,921,266 issued on Mar. 5, 2024, which claims the benefit under 35 USC 119(a) of Korean Patent Application No. 10-2020-0164928 filed on Nov. 30, 2020 in the Korean Intellectual Property Office, the entire disclosures of which are incorporated herein by reference for all purposes.

BACKGROUND

1. Field

[0002] The following description relates to an optical imaging system.

2. Description of Background

[0003] A camera module has been standardly installed in a mobile electronic device including a smartphone.

[0004] In addition, a manner of mounting a plurality of camera modules having different focal lengths in the mobile electronic device in order to indirectly implement an optical zoom effect has been proposed.

[0005] However, such a manner requires the plurality of camera modules for the optical zoom effect, and thus, has a disadvantage that a structure may be complicated.

SUMMARY

[0006] This Summary is provided to introduce a selection of concepts in simplified form that are further described below in the Detailed Description. This Summary is not intended to identify key features or essential features of the claimed subject matter, nor is it intended to be used as an aid in determining the scope of the claimed subject matter.

[0007] An optical imaging system capable of implementing a zoom function by varying a focal length.

[0008] In one general aspect, an optical imaging system includes: a first lens group including a plurality of lenses; a second lens group including a plurality of lenses; a first reflecting member disposed on an object side of the first lens group; and a plurality of second reflecting members arranged between the second lens group and an image sensor, wherein one or both of the first lens group and the second lens group is movable along an optical axis, an interval along the optical axis between the first lens group and the second lens group is varied between a first interval and a second interval smaller than the first interval, and as at least one of the first lens group and the second lens group is moved, one or more of the second reflecting members are moved.

[0009] The first lens group may have negative refractive power, and the second lens group may have positive refractive power.

[0010] The first lens group may include a first lens and a second lens, and the second lens group may include a third lens, a fourth lens, and a fifth lens.

[0011] The first lens may have positive refractive power, and the second lens may have negative refractive power.

[0012] The third lens may have positive refractive power, the fourth lens may have negative refractive power, and the fifth lens may have positive refractive power.

[0013] The optical imaging system may satisfy $f_l/f_h < 0.7$, where f_l is an overall focal length of the optical imaging system when the interval between the first lens group and the second lens group is the first interval, and f_h is an overall focal length of the optical imaging system when the interval between the first lens group and the second lens group is the second interval.

[0014] The optical imaging system may satisfy $BFL_h > 30$ mm, where BFL_h is a distance along the optical axis from an image-side surface of a rearmost lens of the plurality of lenses included in the second lens group to an imaging plane when the interval between the first lens group and the second lens group is the second interval.

[0015] The optical imaging system may satisfy $D1_2/D1_1 < 0.3$, where $D1_1$ is the first interval, and $D1_2$ is the second interval.

[0016] The optical imaging system may satisfy $FOVI/FOV_h > 1.6$, where $FOVI$ is a field of view of the optical imaging system when the interval between the first lens group and the second lens group is the first interval, and FOV_h is a field of view of the optical imaging system when the interval between the first lens group and the second lens group is the second interval.

[0017] The optical imaging system may satisfy $fG1/fG2 < -0.5$, where $fG1$ is a focal length of the first lens group, and $fG2$ is a focal length of the second lens group.

[0018] The second lens group may be configured to be movable along the optical axis such that a distance along the optical axis between the second lens group and the plurality of second reflecting members is varied.

[0019] The plurality of second reflecting members may include a second reflecting member, a third reflecting member, a fourth reflecting member, and a fifth reflecting member sequentially arranged from the second lens group toward the image sensor, the third reflecting member may be configured to be moved, such that an interval between the second reflecting member and the third reflecting member is varied, and the fourth reflecting member may be configured to be moved, such that an interval between the fourth reflecting member and the fifth reflecting member is varied.

[0020] The third reflecting member and the fourth reflecting member may be configured to be moved together.

[0021] The plurality of second reflecting members may include a second reflecting member and a third reflecting member sequentially arranged from the second lens group toward the image sensor.

[0022] The second reflecting member and the third reflecting member may be configured to be moved together.

[0023] The plurality of second reflecting members may include a fourth reflecting member disposed between the third reflecting member and the image sensor.

[0024] The third reflecting member and the fourth reflecting member may be configured to be moved together.

[0025] In another general aspect, an optical imaging system includes: a first lens group; a second lens group; and a reflecting member disposed on an image side of the second lens group to direct light that has passed through the second lens group to an image sensor. One or both of the first lens group and the second lens group are configured to be moved along an optical axis such that an interval between the first lens group and the second lens group along the optical axis is variable between a maximum interval and a minimum interval. The reflecting member is configured to be moved such that a distance that the light that has passed through the second lens group travels between the second lens group and the reflecting member is variable.

[0026] Other features and aspects will be apparent from the following detailed description, the drawings, and the claims.

Description

BRIEF DESCRIPTION OF DRAWINGS

[0027] FIG. 1 is a view illustrating a state in which a first lens group and a second lens group are arranged in a first position in an optical imaging system according to a first example.

[0028] FIG. 2 is graphs having curves representing aberration characteristics of the optical imaging system illustrated in FIG. 1.

[0029] FIG. 3 is a view illustrating a state in which the first lens group and the second lens group are arranged in a second position in the optical imaging system according to the first example.

[0030] FIG. 4 is graphs having curves representing aberration characteristics of the optical imaging system illustrated in FIG. 3.

[0031] FIG. 5 is a view illustrating a state in which a first lens group and a second lens group are arranged in a first position in an optical imaging system according to a second example.

[0032] FIG. 6 is graphs having curves representing aberration characteristics of the optical imaging system illustrated in FIG. 5.

[0033] FIG. 7 is a view illustrating a state in which the first lens group and the second lens group are arranged in a second position in the optical imaging system according to the second example.

[0034] FIG. 8 is graphs having curves representing aberration characteristics of the optical imaging system illustrated in FIG. 7.

[0035] FIG. 9 is a view illustrating a state in which a first lens group and a second lens group are arranged in a first position in an optical imaging system according to a third example.

[0036] FIG. 10 is graphs having curves representing aberration characteristics of the optical imaging system illustrated in FIG. 9.

[0037] FIG. 11 is a view illustrating a state in which the first lens group and the second lens group are arranged in a second position in the optical imaging system according to the third example

[0038] FIG. 12 is graphs having curves representing aberration characteristics of the optical imaging system illustrated in FIG. 11.

[0039] Throughout the drawings and the detailed description, the same reference numerals refer to the same elements. The drawings may not be to scale, and the relative size, proportions, and depictions of elements in the drawings may be exaggerated for clarity, illustration, and convenience.

DETAILED DESCRIPTION

[0040] The following detailed description is provided to assist the reader in gaining a comprehensive understanding of the methods, apparatuses, and/or systems described herein. However, various changes, modifications, and equivalents of the methods, apparatuses, and/or systems described herein will be apparent to one of ordinary skill in the art. The sequences of operations described herein are merely examples, and are not limited to those set forth herein, but may be changed as will be apparent to one of ordinary skill in the art, with the exception of operations necessarily occurring in a certain order. Also, descriptions of functions and constructions that would be well known to one of ordinary skill in the art may be omitted for increased clarity and conciseness.

[0041] The features described herein may be embodied in different forms, and are not to be construed as being limited to the examples described herein. Rather, the examples described herein have been provided so that this disclosure will be thorough and complete, and will fully convey the scope of the disclosure to one of ordinary skill in the art.

[0042] Herein, it is noted that use of the term “may” with respect to an example or embodiment, e.g., as to what an example or embodiment may include or implement, means that at least one example or embodiment exists in which such a feature is included or implemented while all examples and embodiments are not limited thereto.

[0043] Throughout the specification, when an element, such as a layer, region, or substrate, is described as being “on,” “connected to,” or “coupled to” another element, it may be directly “on,” “connected to,” or “coupled to” the other element, or there may be one or more other elements intervening therebetween. In contrast, when an element is described as being “directly on,” “directly connected to,” or “directly coupled to” another element, there can be no other elements intervening therebetween.

[0044] As used herein, the term “and/or” includes any one and any combination of any two or more of the associated listed items.

[0045] Although terms such as “first,” “second,” and “third” may be used herein to describe various members, components, regions, layers, or sections, these members, components, regions, layers, or sections are not to be limited by these terms. Rather, these terms are only used to distinguish one member, component, region, layer, or section from another member, component, region, layer, or section. Thus, a first member, component, region, layer, or section referred to in examples described herein may also be referred to as a second member, component, region, layer, or section without departing from the teachings of the examples.

[0046] Spatially relative terms such as “above,” “upper,” “below,” and “lower” may be used herein for ease of description to describe one element's relationship to another element as illustrated in the figures. Such spatially relative terms are intended to encompass different orientations of the device in use or operation in addition to the orientation depicted in the figures. For example, if the device in the figures is turned over, an element described as being “above” or “upper” relative to another element will then be “below” or “lower” relative to the other element. Thus, the term “above” encompasses both the above and below orientations depending on the spatial orientation of the device. The device may also be oriented in other ways (for example, rotated 90 degrees or at other orientations), and the spatially relative terms used herein are to be interpreted accordingly.

[0047] The terminology used herein is for describing various examples only, and is not to be used to limit the disclosure. The articles “a,” “an,” and “the” are intended to include the plural forms as well, unless the context clearly indicates otherwise. The terms “comprises,” “includes,” and “has” specify the presence of stated features, numbers, operations, members, elements, and/or combinations thereof, but do not preclude the presence or addition of one or more other features, numbers, operations, members, elements, and/or combinations thereof.

[0048] Due to manufacturing techniques and/or tolerances, variations of the shapes illustrated in the drawings may occur. Thus, the examples described herein are not limited to the specific shapes illustrated in the drawings, but include changes in shape that occur during manufacturing.

[0049] The features of the examples described herein may be combined in various ways as will be apparent after an understanding of the disclosure of this application. Further, although the examples described herein have a variety of configurations, other configurations are possible as will be apparent after an understanding of the disclosure of this application.

[0050] The drawings may not be to scale, and the relative sizes, proportions, and depiction of elements in the drawings may be exaggerated for clarity, illustration, and convenience.

[0051] Hereinafter, examples will be described in detail with reference to the accompanying drawings.

[0052] In the drawings, thicknesses, sizes, and shapes of lenses have been slightly exaggerated for convenience of explanation. Particularly, shapes of spherical surfaces or aspherical surfaces illustrated in the drawings are illustrated by way of example. That is, the shapes of the spherical surfaces or the aspherical surfaces are not limited to those illustrated in the drawings.

[0053] An optical imaging system according to various examples may be mounted in a mobile electronic device. For example, the optical imaging system may be one component of a camera module mounted in the mobile electronic device. The mobile electronic device may be a portable electronic device such as a mobile communications terminal, a smartphone, or a tablet personal computer (PC).

[0054] The optical imaging system may include a plurality of lenses. The plurality of lenses may be arranged to be spaced apart from each other by preset distances.

[0055] As an example, the optical imaging system may include at least five lenses.

[0056] A first lens (or a foremost lens) refers to a lens closest to an object side (or a first reflecting member), while a last lens (or a rearmost lens) refers to a lens closest to an image sensor.

[0057] In addition, a first surface of each lens refers to a surface thereof close to the object side (or an object-side surface) and a second surface of each lens refers to a surface thereof close to an image side (or an image-side surface). Further, in the present specification, all numerical values of radii of curvature, thicknesses, distances, and the like, of lenses are indicated by millimeters (mm), and a field of view (FOV) is indicated by degrees.

[0058] Further, in a description for a shape of each of the lenses, the meaning that one surface of a lens is convex is that a paraxial region portion of a corresponding surface is convex, and the meaning that one surface of a lens is concave is that a paraxial region portion of a corresponding surface is concave. Therefore, although it is described that one surface of a lens is convex, an edge portion of the lens may be concave. Likewise, although it is described that one surface of a lens is concave, an edge portion of the lens may be convex.

[0059] A paraxial region refers to a very narrow region in the vicinity of an optical axis.

[0060] An optical imaging system according to the various embodiments may include at least five lenses.

[0061] For example, the optical imaging system may include a first lens, a second lens, a third lens, a fourth lens, and a fifth lens sequentially arranged from the object side (or the first reflecting member).

[0062] The optical imaging system may include six or more lenses, if necessary. In addition, the optical imaging system may further include an image sensor converting an image of a subject incident on the image sensor into an electrical signal.

[0063] The optical imaging system may further include a plurality of reflecting members each having a reflecting surface changing a path of light. As an example, each of the plurality of reflecting members may be a mirror or a prism.

[0064] Any one of the reflecting members may be disposed in front of the plurality of lenses. As an example, a first reflecting member may be disposed in front of the first lens (that is, closer to the object side than the first lens is). The other reflecting members may be arranged behind the plurality of lenses. As an example, the other reflecting members may be arranged between the fifth lens and the image sensor.

[0065] The plurality of reflecting members may be arranged between the plurality of lenses and the image sensor to refract the light several times, resulting in an elongated path of light in a relatively narrow space.

[0066] Therefore, the optical imaging system may be miniaturized and have a great focal length.

[0067] Further, the optical imaging system may further include an infrared cut-off filter (hereinafter, referred to as a filter) cutting off infrared rays. The filter may be disposed between a reflecting member disposed closest to the image sensor among the plurality of reflecting members and the image sensor.

[0068] In the optical imaging system according to the various examples, all of the lenses may be formed of plastic materials.

[0069] All of the lenses may have an aspherical surface. For example, each of the first to fifth lenses may have at least one aspherical surface.

[0070] That is, at least one of first and second surfaces of all of the first to fifth lenses may be aspherical. Here, the aspherical surfaces of the first to fifth lenses may be represented by the following Equation 1:

[00001] [Equation1]
$$Z = \frac{cY^2}{1 + \sqrt{1 - (1 + K)c^2Y^2}} + AY^4 + BY^6 + CY^8 + DY^{10} + EY^{12} \text{ .Math.}$$

[0071] In equation 1, c is a curvature (an inverse of a radius of curvature) of a lens, K is a conic constant, and Y is a distance from a certain point on an aspherical surface of the lens to an optical axis in a direction perpendicular to the optical axis. In addition, constants A to E are aspherical coefficients. In addition, Z is a distance (SAG) between the certain point on the aspherical surface of the lens at the distance Y and a tangential plane meeting the apex of the aspherical surface of the lens.

[0072] The first to fifth lenses may have positive refractive power/negative refractive power/positive refractive power/negative refractive power/positive refractive power sequentially from the object side.

[0073] The optical imaging system according to the various examples may satisfy at least one of the following Conditional Expressions:

[00002] $f_l / f_h < 0.7$; [ConditionalExpression1] $BFL_h > 30\text{mm}$; [ConditionalExpression2]

$D1_2 / D1_1 < 0.3$; [ConditionalExpression3]

$FOV_l / FOV_h > 1.6$; and [ConditionalExpression4]

$fG1 / fG2 < -0.5$. [ConditionalExpression5]

[0074] In the conditional expressions, f_l is an overall focal length of the optical imaging system in a first position, and f_h is an overall focal length of the optical imaging system in a second position. The first position refers to a case in which a first lens group and a second lens group are spaced apart from each other by a first interval, and the second position refers to a case in which the first lens group and the second lens group are spaced apart from each other by a second interval. The second interval may be narrower than the first interval. The optical imaging system may have a greater overall focal length in the second position than in the first position.

[0075] BFL_h is a distance on the optical axis from an image-side surface of the last lens (for example, the fifth lens) to an imaging plane of the image sensor in the second position.

[0076] $D1_1$ is a distance on the optical axis between the first lens group and the second lens group in the first position, and $D1_2$ is a distance on the optical axis between the first lens group and the second lens group in the second position.

[0077] FOV_l is a field of view of the optical imaging system in the first position, and FOV_h is a field of view of the optical imaging system in the second position. The optical imaging system may have a narrower field of view in the second position than in the first position.

[0078] $fG1$ is a focal length of the first lens group among the plurality of lenses of the optical imaging system, and $fG2$ is a focal length of the second lens group among the plurality of lenses of the optical imaging system.

[0079] The optical imaging system according to the various examples may include a plurality of lens groups. As an example, the optical imaging system may include the first lens group and the second lens group. Each of the first lens group and the second lens group may include a plurality of lenses. The first lens group may be disposed in front of the second lens group.

[0080] The first lens group may include a first lens and a second lens. The first lens may have positive refractive power, and a first surface and a second surface of the first lens may be convex. The second lens may have negative refractive power, and a first surface and a second surface of the second lens may be concave. Alternatively, a first surface of the second lens may be concave, and a second surface of the second lens may be convex.

[0081] The first lens group may have negative refractive power as a whole.

[0082] The second lens group may include a third lens, a fourth lens, and a fifth lens. The third lens may have positive refractive power, and a first surface and a second surface of the third lens may be convex. The fourth lens may have negative refractive power, and a first surface and a second surface of the fourth lens may be concave. The fifth lens may have positive refractive power, and a first surface and a second surface of the fifth lens may be convex.

[0083] The second lens group may have positive refractive power as a whole.

[0084] At least one of the first lens group and the second lens group may be moved to change the overall focal length of the optical imaging system. For example, an interval between the first lens group and the second lens group may be varied. As an example, the first lens group may be fixedly disposed, and the second lens group may be movably disposed. In addition, an interval between the second lens group and the image sensor may also be varied.

[0085] At least a portion of the plurality of reflecting members arranged between the second lens group and the image sensor may be moved. Since the interval between the second lens group and the image sensor may be varied by moving at least a portion of the plurality of reflecting members, a focus may be easily formed on the image sensor even though a focal length of the optical imaging system is changed due to the movement of the second lens group.

[0086] Therefore, the optical imaging system may have an optical zoom function and/or a focusing function.

[0087] The optical image system may have a feature of a telephoto lens having a relatively narrow field of view and great focal length.

[0088] An optical imaging system according to a first example will be described with reference to FIGS. 1 through 4.

[0089] FIG. 1 is a view illustrating a state in which a first lens group and a second lens group are arranged in a first position, and FIG. 3 is a view illustrating a state in which the first lens group and the second lens group are arranged in a second position.

[0090] The optical imaging system according to the first example may include a first lens group G1 and a second lens group G2.

[0091] The first lens group G1 may include a first lens 110 and a second lens 120, and the second lens group G2 may include a third lens 130, a fourth lens 140, and a fifth lens 150. In addition, the optical imaging system may further include a filter and an image sensor 160. In addition, a stop may be disposed between the first lens group G1 and the second lens group G2. For example, the stop may be disposed between the second lens 120 and the third lens 130.

[0092] In addition, the optical imaging system may further include a first reflecting member R1 disposed in front of the first lens 110 and having a reflecting surface changing a path of light. In the first example, the first reflecting member R1 may be a prism, but may also be a mirror.

[0093] A case in which the prism has a flat panel shape is illustrated in FIGS. 1 and 3, but the prism may actually have a triangular prismatic shape.

[0094] Light incident on the first reflecting member R1 may be refracted by the first reflecting member R1 and then pass through the first lens group G1 and the second lens group G2. For example, light incident on the first reflecting member R1 along a first optical axis may be refracted toward a second optical axis perpendicular to the first optical axis.

[0095] In addition, the optical imaging system may further include a second reflecting member R2, a third reflecting member R3, a fourth reflecting member R4, and a fifth reflecting member R5 arranged behind the fifth lens 150, and each having a reflecting surface changing a path of the light. The second reflecting member R2, the third reflecting member R3, the fourth reflecting member R4, and the fifth reflecting member R5 may be arranged between the fifth lens 150 and the image sensor 160.

[0096] In the first example, the second reflecting member R2, the third reflecting member R3, the fourth reflecting member R4, and the fifth reflecting member R5 may be mirrors, but may also be prisms.

[0097] The light passing through the first lens group G1 and the second lens group G2 may be refracted by the second reflecting member R2. For example, the light passing through the first lens group G1 and the second lens group G2 may be refracted toward a third optical axis perpendicular to the second optical axis by the second reflecting member R2.

[0098] The light refracted by the second reflecting member R2 may be refracted again by the third reflecting member R3. For example, the light refracted toward the third optical axis by the second

reflecting member **R2** may be refracted toward a fourth optical axis perpendicular to the third optical axis by the third reflecting member **R3**.

[0099] The light refracted by the third reflecting member **R3** may be refracted again by the fourth reflecting member **R4**. For example, the light refracted toward the fourth optical axis by the third reflecting member **R3** may be refracted toward a fifth optical axis perpendicular to the fourth optical axis by the fourth reflecting member **R4**.

[0100] The light refracted by the fourth reflecting member **R4** may be refracted again by the fifth reflecting member **R5**. For example, the light refracted toward the fifth optical axis by the fourth reflecting member **R4** may be refracted toward a sixth optical axis perpendicular to the fifth optical axis by the fifth reflecting member **R5**, and be then received by the image sensor **160**.

[0101] The filter may be disposed between the fifth reflecting member **R5** and the image sensor **160**.

[0102] At least one of the first lens group **G1** and the second lens group **G2** may be moved in order to change an overall focal length of the optical imaging system. As an example, the second lens group **G2** may be moved along the optical axis of the optical imaging system, such that an interval between the first lens group **G1** and the second lens group **G2** may be varied and an interval between the second lens group **G2** and the second reflecting member **R2** may be varied. In addition, an interval between the second lens group **G2** and the image sensor **160** may be varied. Here, the ‘interval’ may refer to a distance on the optical axis between two members.

[0103] At least two of the second reflecting member **R2**, the third reflecting member **R3**, the fourth reflecting member **R4**, and the fifth reflecting member **R5** may be moved. For example, the third reflecting member **R3** may be moved, such that an interval between the second reflecting member **R2** and the third reflecting member **R3** may be varied. In addition, the fourth reflecting member **R4** may be moved, such that an interval between the fourth reflecting member **R4** and the fifth reflecting member **R5** may be varied. The third reflecting member **R3** and the fourth reflecting member **R4** may be moved together.

[0104] A length of the path of the light may be changed by the third reflecting member **R3** and the fourth reflecting member **R4**.

[0105] In the present example, the first reflecting member **R1**, the second reflecting member **R2**, and the fifth reflecting member **R5** may be fixed members whose positions are fixed, and the third reflecting member **R3** and the fourth reflecting member **R4** may be movable members whose positions may be varied.

[0106] Characteristics (radii of curvature, thicknesses of lenses or distances between the lenses, refractive indices, Abbe numbers, and focal lengths) of each lens are represented in Table 1.

TABLE-US-00001

TABLE 1	Surface	Radius of Curvature or Distance	Thickness	Refractive Index	Abbe Number	Focal Length	No.	Remark
S1	First Reflecting	Infinity	2.212	1.717	29.5	Member		
S2	Infinity	2.212	1.717	29.5				
S3	Infinity	1.942						
S4	First Lens	800.250	1.902	1.671	19.2	25.916	S5	
		-17.991	0.850				S6	
S6	Second Lens	-5.276	1.950	1.544	56	-7.268	S7	
		18.158					D1	
S8	Third Lens	800.000	3.000	1.544	56	22.393	S9	
		-12.410	0.350				S10	
S10	Fourth Lens	-13.506	1.443	1.595	31.1		S11	
		-9.176					S12	
S12	Fifth Lens	8.586	2.571	1.544	56	7.851	S13	
		-7.667					D2	
S14	Second Reflecting	Infinity	0.000				Member	
S15	Infinity	D3					S16	
S16	Third Reflecting	Infinity	0.000				Member	
S17	Infinity	7.500					S18	
S18	Fourth Reflecting	Infinity	0.000				Member	
S19	Infinity	D4					S20	
S20	Fifth Reflecting	Infinity	0.000				Member	
S21	Infinity	5.000					S22	
S22	Imaging Plane	Infinity						

TABLE-US-00002

TABLE 2	First Second	Position	Position	D1	D2	D3	D4
		6	1.5	4.5	9	8.125	15.821
		8.125	15.821				

[0107] **D1** is a distance on the optical axis between the second lens **120** and the third lens **130**, **D2** is a distance on the optical axis between the fifth lens **150** and the second reflecting member **R2**, **D3** is a distance on the optical axis between the second reflecting member **R2** and the third reflecting member **R3**, and **D4** is a distance on the optical axis between the fourth reflecting member **R4** and the fifth reflecting member **R5**.

[0108] An interval between the first lens group G1 and the second lens group G2 in the first position may be different from that between the first lens group G1 and the second lens group G2 in the second position. For example, the first lens group G1 and the second lens group G2 may be located relatively distant from each other in the first position, and may be located relatively close to each other in the second position.

[0109] The overall focal length of the optical imaging system may be changed by varying the interval between the first lens group G1 and the second lens group G2.

[0110] In addition, a length of the path of the light between the second lens group G2 and the image sensor 160 may be changed by moving some of a plurality of reflecting members arranged between the second lens group G2 and the image sensor 160. For example, positions of the third reflecting member R3 and the fourth reflecting member R4 in the first position may be different from positions of the third reflecting member R3 and the fourth reflecting member R4 in the second position.

[0111] In the first position, an overall focal length f_l of the optical imaging system may be 16 mm, Fno_I may be 5.16, $FOVI$ may be 10.1° , BFL_I may be 33.250 mm, TTL_I may be 51.636 mm, and $PTTL_I$ may be 58.003 mm.

[0112] In the second position, an overall focal length f_h of the optical imaging system may be 32 mm, Fno_h may be 8.17, $FOVh$ may be 5° , BFL_h may be 53.142 mm, TTL_h may be 67.028 mm, and $PTTL_h$ may be 73.395 mm.

[0113] Fno_I is an F-number of the optical imaging system in the first position, and Fno_h is an F-number of the optical imaging system in the second position.

[0114] $FOVI$ is a field of view of the optical imaging system in the first position, and $FOVh$ is a field of view of the optical imaging system in the second position.

[0115] BFL_I is a distance on the optical axis from an image-side surface of the fifth lens to an imaging plane of the image sensor in the first position, and BFL_h is a distance on the optical axis from the image-side surface of the fifth lens to the imaging plane of the image sensor in the second position.

[0116] TTL_I is a distance on the optical axis from an object-side surface of the first lens to the imaging plane of the image sensor in the first position, and TTL_h is a distance on the optical axis from the object-side surface of the first lens to the imaging plane of the image sensor in the second position.

[0117] $PTTL_I$ is a distance on the optical axis from an object-side surface (or an incident surface) of the first reflecting member to the imaging plane of the image sensor in the first position, and $PTTL_h$ is a distance on the optical axis from the object-side surface (or the incident surface) of the first reflecting member to the imaging plane of the image sensor in the second position.

[0118] In the first example, the first lens group G1 may have negative refractive power as a whole. A focal length of the first lens group G1 may be -10.766 mm.

[0119] The second lens group G2 may have positive refractive power as a whole. A focal length of the second lens group G2 may be 13.393 mm.

[0120] The first lens 110 may have positive refractive power, and a first surface and a second surface thereof may be convex.

[0121] The second lens 120 may have negative refractive power, and a first surface and a second surface thereof may be concave.

[0122] The third lens 130 may have positive refractive power, and a first surface and a second surface thereof may be convex.

[0123] The fourth lens 140 may have negative refractive power, and a first surface and a second surface thereof may be concave.

[0124] The fifth lens 150 may have positive refractive power, and a first surface and a second surface thereof may be convex.

[0125] Respective surfaces of the first to fifth lenses 110 to 150 may have aspherical coefficients as

represented in Table 3. For example, all of object-side surfaces and image-side surfaces of the first to fifth lenses **110** to **150** may be aspherical.

TABLE-US-00003 TABLE 3 4th-Order 6th-Order 8th-Order 10th-Order 12th-Order Conic											
Coefficient	Coefficient	Coefficient	Coefficient	Coefficient	Constant (K)	(A)	(B)	(C)	(D)	(E)	S4 0
7.223E-04	1.151E-05	4.670E-06	-6.291E-07	2.750E-08	S5 0	4.744E-04	4.096E-05				
-1.174E-06	1.261E-07	-7.009E-08	S6 0	6.629E-04	2.838E-05	2.732E-07	-3.917E-07				
-4.631E-08	S7 0	2.159E-05	-2.146E-05	4.083E-06	-3.519E-07	1.032E-08	S8 0				
-4.440E-04	-9.538E-06	-2.692E-06	3.367E-07	-2.851E-08	S9 0	1.363E-04	-1.509E-05				
1.587E-06	-4.646E-07	-7.799E-09	S10 0	-1.060E-04	-5.914E-06	-2.221E-06	-3.493E-07				
-2.486E-09	S11 0	-4.393E-05	-1.687E-05	-4.610E-06	5.883E-08	1.951E-09	S12 0				
4.939E-06	-3.569E-06	-8.002E-07	1.115E-07	-1.376E-09	S13 0	1.438E-04	2.262E-05				
-7.418E-07	2.228E-07	-2.247E-09									

[0126] In addition, the optical imaging system configured as described above may have aberration characteristics illustrated in FIGS. **2** and **4**.

[0127] An optical imaging system according to a second example will be described with reference to FIGS. **5** through **8**.

[0128] FIG. **5** is a view illustrating a state in which a first lens group and a second lens group are arranged in a first position, and FIG. **7** is a view illustrating a state in which the first lens group and the second lens group are arranged in a second position.

[0129] The optical imaging system according to the second example may include a first lens group **G1** and a second lens group **G2**.

[0130] The first lens group **G1** may include a first lens **210** and a second lens **220**, and the second lens group **G2** may include a third lens **230**, a fourth lens **240**, and a fifth lens **250**. In addition, the optical imaging system may further include a filter and an image sensor **260**. In addition, a stop may be disposed between the first lens group **G1** and the second lens group **G2**. For example, the stop may be disposed between the second lens **220** and the third lens **230**.

[0131] In addition, the optical imaging system may further include a first reflecting member **R1** disposed in front of the first lens **210** and having a reflecting surface changing a path of light. In the second example, the first reflecting member **R1** may be a prism, but may also be a mirror.

[0132] A case in which the prism has a flat panel shape is illustrated in FIGS. **5** and **7**, but the prism may actually have a triangular prismatic shape.

[0133] Light incident on the first reflecting member **R1** may be refracted by the first reflecting member **R1** and then pass through the first lens group **G1** and the second lens group **G2**. For example, light incident on the first reflecting member **R1** along a first optical axis may be refracted toward a second optical axis perpendicular to the first optical axis.

[0134] In addition, the optical imaging system may further include a second reflecting member **R2** and a third reflecting member **R3** arranged behind the fifth lens **250** and each having a reflecting surface changing a path of light. The second reflecting member **R2** and the third reflecting member **R3** may be arranged between the fifth lens **250** and the image sensor **260**.

[0135] In the second example, the second reflecting member **R2** and the third reflecting member **R3** may be mirrors, but may also be prisms.

[0136] The light passing through the first lens group **G1** and the second lens group **G2** may be refracted by the second reflecting member **R2**. For example, the light passing through the first lens group **G1** and the second lens group **G2** may be refracted toward a third optical axis perpendicular to the second optical axis by the second reflecting member **R2**.

[0137] The light refracted by the second reflecting member **R2** may be refracted again by the third reflecting member **R3**. For example, the light refracted toward the third optical axis by the second reflecting member **R2** may be refracted toward a fourth optical axis perpendicular to the third optical axis by the third reflecting member **R3**, and be then received by the image sensor **260**.

[0138] The filter may be disposed between the third reflecting member **R3** and the image sensor

260.

[0139] At least one of the first lens group G1 and the second lens group G2 may be moved in order to change an overall focal length of the optical imaging system. As an example, the second lens group G2 may be moved along the optical axis, such that an interval between the first lens group G1 and the second lens group G2 may be varied and an interval between the second lens group G2 and the second reflecting member R2 may be varied. In addition, an interval between the second lens group G2 and the image sensor 260 may be varied. Here, the 'interval' may refer to a distance on the optical axis between two members.

[0140] The second reflecting member R2 and the third reflecting member R3 may be moved. For example, the second reflecting member R2 and the third reflecting member R3 may be moved together, such that an interval between the second lens group G2 and the second reflecting member R2 and an interval between the third reflecting member R3 and the image sensor 260 may be varied.

[0141] In the present example, the first reflecting member R1 may be a fixed member whose position is fixed, and the second reflecting member R2 and the third reflecting member R3 may be movable members whose positions may be varied.

[0142] Characteristics (radii of curvature, thicknesses of lenses or distances between the lenses, refractive indices, Abbe numbers, and focal lengths) of each lens are represented in Table 4.

TABLE-US-00004 TABLE 4 Surface Radius of Thickness Refractive Abbe Focal No. Remark
Curvature or Distance Index Number Length S1 First Reflecting Infinity 2.212 1.717 29.5 Member
S2 Infinity 2.212 1.717 29.5 S3 Infinity 1.942 S4 First Lens 34.849 2.750 1.671 19.2 20.781 S5
-22.988 0.522 S6 Second Lens -4.680 1.950 1.573 37 -9.289 S7 -42.664 D1 S8 Third Lens
13.422 0.955 1.544 56 13.838 S9 -16.880 0.256 S10 Fourth Lens -7.908 1.129 1.586 32.6 -7.742
S11 11.378 0.420 S12 Fifth Lens 11.995 2.445 1.544 56 8.283 S13 -6.746 D2 S14 Second
Reflecting Infinity 0.000 Member S15 Infinity 7.455 S16 Third Reflecting Infinity 0.000 Member
S17 Infinity D3 S18 Imaging Plane Infinity

TABLE-US-00005 TABLE 5 First Second Position Position D1 8.339 0.689 D2 4 11.65 D3 11.544
11.562

[0143] D1 is a distance on the optical axis between the second lens 220 and the third lens 230, D2 is a distance on the optical axis between the fifth lens 250 and the second reflecting member R2, and D3 is a distance on the optical axis between the third reflecting member R3 and the image sensor 260.

[0144] An interval between the first lens group G1 and the second lens group G2 in the first position may be different from that between the first lens group G1 and the second lens group G2 in the second position. For example, the first lens group G1 and the second lens group G2 may be located relatively distant from each other in the first position, and may be located relatively close to each other in the second position.

[0145] The overall focal length of the optical imaging system may be changed by varying the interval between the first lens group G1 and the second lens group G2.

[0146] In addition, a length of the path of the light between the second lens group G2 and the image sensor 260 may be changed by moving a plurality of reflecting members arranged between the second lens group G2 and the image sensor 260. For example, positions of the second reflecting member R2 and the third reflecting member R3 in the first position may be different from positions of the second reflecting member R2 and the third reflecting member R3 in the second position.

[0147] In the first position, an overall focal length fl of the optical imaging system may be 13.9 mm, Fno_I may be 3.88, FOVI may be 11.5°, BFL_I may be 23,000 mm, TTL_I may be 41.767 mm, and PTTL_I may be 48.134 mm.

[0148] In the second position, an overall focal length fh of the optical imaging system may be 24.6 mm, Fno_h may be 4.97, FOVh may be 6.5°, BFL_h may be 30.667 mm, TTL_h may be 41.784 mm, and PTTL_h may be 48.151 mm.

[0149] In the second example, the first lens group **G1** may have negative refractive power as a whole. A focal length of the first lens group **G1** may be -18.451 mm.

[0150] The second lens group **G2** may have positive refractive power as a whole. A focal length of the second lens group **G2** may be 13.252 mm.

[0151] The first lens **210** may have positive refractive power, and a first surface and a second surface thereof may be convex.

[0152] The second lens **220** may have negative refractive power, and a first surface thereof may be concave and a second surface thereof may be convex.

[0153] The third lens **230** may have positive refractive power, and a first surface and a second surface thereof may be convex.

[0154] The fourth lens **240** may have negative refractive power, and a first surface and a second surface thereof may be concave.

[0155] The fifth lens **250** may have positive refractive power, and a first surface and a second surface thereof may be convex.

[0156] Respective surfaces of the first to fifth lenses **210** to **250** may have aspherical coefficients as represented in Table 6. For example, all of object-side surfaces and image-side surfaces of the first to fifth lenses **210** to **250** may be aspherical.

TABLE-US-00006 TABLE 6 4th-Order 6th-Order 8th-Order 10th-Order 12th-Order Conic Coefficient Coefficient Coefficient Coefficient Coefficient Constant (K) (A) (B) (C) (D) (E) S4 0 1.014E-03 3.675E-05 1.181E-06 1.307E-08 $-1.065E-09$ S5 0 3.881E-04 1.019E-04 6.765E-06 3.148E-08 3.894E-08 S6 0 9.742E-04 9.341E-05 6.824E-06 $-6.105E-07$ $-7.356E-19$ S7 0 6.175E-04 $-1.853E-05$ 1.236E-06 $-2.471E-07$ 4.069E-19 S8 0 $-4.097E-04$ $-2.042E-05$ $-5.888E-06$ 6.223E-08 $-3.780E-09$ S9 0 7.849E-05 $-5.173E-06$ 8.819E-07 $-5.267E-07$ $-1.913E-18$ S10 0 1.386E-04 1.333E-05 3.180E-07 $-3.562E-07$ 4.974E-19 S11 0 $-1.606E-04$ $-2.622E-05$ $-5.040E-06$ 7.900E-08 $-2.235E-18$ S12 0 $-2.973E-05$ $-5.570E-06$ $-1.102E-06$ 8.573E-08 $-3.864E-09$ S13 0 2.770E-04 1.974E-05 $-1.139E-06$ 2.339E-07 $-5.720E-09$

[0157] In addition, the optical imaging system configured as described above may have aberration characteristics illustrated in FIGS. 6 and 8.

[0158] An optical imaging system according to a third example will be described with reference to FIGS. 9 through 12.

[0159] FIG. 9 is a view illustrating a state in which a first lens group and a second lens group are arranged in a first position, and FIG. 11 is a view illustrating a state in which the first lens group and the second lens group are arranged in a second position.

[0160] The optical imaging system according to the third example may include a first lens group **G1** and a second lens group **G2**.

[0161] The first lens group **G1** may include a first lens **310** and a second lens **320**, and the second lens group **G2** may include a third lens **330**, a fourth lens **340**, and a fifth lens **350**. In addition, the optical imaging system may further include a filter and an image sensor **360**. In addition, a stop may be disposed between the first lens group **G1** and the second lens group **G2**. For example, the stop may be disposed between the second lens **320** and the third lens **330**.

[0162] In addition, the optical imaging system may further include a first reflecting member **R1** disposed in front of the first lens **310** and having a reflecting surface changing a path of light. In the third example, the first reflecting member **R1** may be a prism, but may also be a mirror.

[0163] A case in which the prism has a flat panel shape is illustrated in FIGS. 9 and 11, but the prism may actually have a triangular prismatic shape.

[0164] Light incident on the first reflecting member **R1** may be refracted by the first reflecting member **R1** and then pass through the first lens group **G1** and the second lens group **G2**. For example, light incident on the first reflecting member **R1** along a first optical axis may be refracted toward a second optical axis perpendicular to the first optical axis.

[0165] In addition, the optical imaging system may further include a second reflecting member R2, a third reflecting member R3, and a fourth reflecting member R4 arranged behind the fifth lens 350 and each having a reflecting surface changing a path of light. The second reflecting member R2, the third reflecting member R3, and the fourth reflecting member R4 may be arranged between the fifth lens 350 and the image sensor 360.

[0166] In the third example, the second reflecting member R2, the third reflecting member R3, and the fourth reflecting member R4 may be mirrors, but may also be prisms.

[0167] The light passing through the first lens group G1 and the second lens group G2 may be refracted by the second reflecting member R2. For example, the light passing through the first lens group G1 and the second lens group G2 may be refracted toward a third optical axis perpendicular to the second optical axis by the second reflecting member R2.

[0168] The light refracted by the second reflecting member R2 may be refracted again by the third reflecting member R3. For example, the light refracted toward the third optical axis by the second reflecting member R2 may be refracted toward a fourth optical axis perpendicular to the third optical axis by the third reflecting member R3.

[0169] The light refracted by the third reflecting member R3 may be refracted again by the fourth reflecting member R4. For example, the light refracted toward the fourth optical axis by the third reflecting member R3 may be refracted toward a fifth optical axis perpendicular to the fourth optical axis by the fourth reflecting member R4, and be then received by the image sensor 360.

[0170] The filter may be disposed between the fourth reflecting member R4 and the image sensor 360.

[0171] At least one of the first lens group G1 and the second lens group G2 may be moved in order to change an overall focal length of the optical imaging system. As an example, the second lens group G2 may be moved along the optical axis, such that an interval between the first lens group G1 and the second lens group G2 may be varied and an interval between the second lens group G2 and the second reflecting member R2 may be varied. In addition, an interval between the second lens group G2 and the image sensor 360 may be varied. Here, the 'interval' may refer to a distance on the optical axis between two members.

[0172] At least two of the second reflecting member R2, the third reflecting member R3, and the fourth reflecting member R4 may be moved. For example, the third reflecting member R3 may be moved, such that an interval between the second reflecting member R2 and the third reflecting member R3 may be varied. In addition, the fourth reflecting member R4 may be moved, such that an interval between the fourth reflecting member R4 and the image sensor 360 may be varied. The third reflecting member R3 and the fourth reflecting member R4 may be moved together.

[0173] In the present example, the first reflecting member R1 and the second reflecting member R2 may be fixed members whose positions are fixed, and the third reflecting member R3 and the fourth reflecting member R4 may be movable members whose positions may be varied.

[0174] Characteristics (radii of curvature, thicknesses of lenses or distances between the lenses, refractive indices, Abbe numbers, and focal lengths) of each lens are represented in Table 7.

TABLE-US-00007 TABLE 7 Surface Radius of Thickness Refractive Abbe Focal No. Remark													
Curvature or Distance	Index	Number	Length	S1	First Reflecting	Infinity	2.212	1.717	29.5	Member			
S2	Infinity	2.212	1.717	29.5	S3	Infinity	1.942	S4	First Lens	53.925	2.750	1.671	19.2 24.611 S5
-23.745	0.662	S6	Second Lens	-4.822	1.871	1.562	42.4	-9.584	S7	-50.14916	D1	S8	Third Lens
12.161	0.967	1.544	56	17.207	S9	-40.279	0.275	S10	Fourth Lens	-10.348	1.288	1.593	31.7 -8.185
S11	9.691	0.330	S12	Fifth Lens	9.863	2.556	1.544	56	7.730	S13	-6.711	D2	S14
Infinity	0.000	Member	S15	Infinity	D3	S16	Third Reflecting	Infinity	0.000	Member	S17	Infinity	
7.15	S18	Fourth Reflecting	Infinity	0.000	Member	S19	Infinity	D4	S20	Imaging Plane	Infinity		

TABLE-US-00008 TABLE 8 First Second Position Position													
D1	9.520	0.7424	D2	4	12.777	D3	7.448	7.691	D4	11.484	11.726		

[0175] D1 is a distance on the optical axis between the second lens 320 and the third lens 330, D2

[0189] In addition, the optical imaging system configured as described above may have aberration characteristics illustrated in FIGS. 10 and 12.

[0190] As set forth above, the optical imaging system according to the various examples may implement a zoom function by varying a focal length.

[0191] While this disclosure includes specific examples, it will be apparent to one of ordinary skill in the art that various changes in form and details may be made in these examples without departing from the spirit and scope of the claims and their equivalents. The examples described herein are to be considered in a descriptive sense only, and not for purposes of limitation. Descriptions of features or aspects in each example are to be considered as being applicable to similar features or aspects in other examples. Suitable results may be achieved if the described techniques are performed to have a different order, and/or if components in a described system, architecture, device, or circuit are combined in a different manner, and/or replaced or supplemented by other components or their equivalents. Therefore, the scope of the disclosure is defined not by the detailed description, but by the claims and their equivalents, and all variations within the scope of the claims and their equivalents are to be construed as being included in the disclosure.

Claims

1. An optical imaging system comprising: a first lens group comprising a first lens having a convex object-side surface in a paraxial region and a second lens; a second lens group comprising a third lens, a fourth lens, and a fifth lens, the optical imaging system comprising a total number of five lenses; and a plurality of reflecting members disposed on an image side of the second lens group and configured to direct light that has passed through the second lens group to an image sensor, wherein the second lens group are configured to be moved along an optical axis such that an interval between the first lens group and the second lens group along the optical axis is variable between a maximum interval and a minimum interval, and two of the plurality of reflecting members are configured to be moved together in a same direction such that a distance that the light that has passed through the second lens group travels between the second lens group and the image sensor is variable.
2. The optical imaging system of claim 1, wherein the first lens group has negative refractive power, and the second lens group has positive refractive power.
3. The optical imaging system of claim 1, wherein the first lens has positive refractive power, and the second lens has negative refractive power.
4. The optical imaging system of claim 1, wherein the third lens has positive refractive power, the fourth lens has negative refractive power, and the fifth lens has positive refractive power.
5. The optical imaging system of claim 1, wherein $f_l/f_h < 0.7$, where f_l is an overall focal length of the optical imaging system when the interval between the first lens group and the second lens group is the maximum interval, and f_h is an overall focal length of the optical imaging system when the interval between the first lens group and the second lens group is the minimum interval.
6. The optical imaging system of claim 5, wherein $BFL_h > 30$ mm, where BFL_h is a distance along the optical axis from an image-side surface of the fifth lens to an imaging plane when the interval between the first lens group and the second lens group is the minimum interval.
7. The optical imaging system of claim 1, wherein $D1_2/D1_1 < 0.3$, where $D1_1$ is the first interval, and $D1_2$ is the minimum interval.
8. The optical imaging system of claim 1, wherein $FOVI/FOVh > 1.6$, where $FOVI$ is a field of view of the optical imaging system when the interval between the first lens group and the second lens group is the first interval, and $FOVh$ is a field of view of the optical imaging system when the interval between the first lens group and the second lens group is the minimum interval.
9. The optical imaging system of claim 1, wherein $fG1/fG2 < -0.5$, where $fG1$ is a focal length of the first lens group, and $fG2$ is a focal length of the second lens group.

10. The optical imaging system of claim 1, wherein the plurality of reflecting members comprise a first reflecting member, a second reflecting member, a third reflecting member, and a fourth reflecting member sequentially arranged from the second lens group toward the image sensor, the second reflecting member is configured to be moved, such that an interval between the first reflecting member and the second reflecting member is varied, and the third reflecting member is configured to be moved, such that an interval between the third reflecting member and the fourth reflecting member is varied.

11. The optical imaging system of claim 10, wherein the second reflecting member and the third reflecting member are configured to be moved together.

12. The optical imaging system of claim 1, wherein the plurality of reflecting members comprise a first reflecting member, a second reflecting member, and a third reflecting member sequentially arranged from the second lens group toward the image sensor, and wherein the second reflecting member and the third reflecting member are configured to be moved together.

13. The optical imaging system of claim 1, wherein the plurality of reflecting members includes at least one fixed member and two movable members, and the two movable members of the plurality of reflecting members are configured to be moved together in a same direction.

14. The optical imaging system of claim 1, wherein each lens in the first lens group and the second lens group has at least one aspherical surface.
